

STM32Axis Development Board

Short Design Description

1. Problem Statement

The objective is to design a compact and functional STM32-based evaluation and development board that provides the essential facilities required for programming, debugging, peripheral interfacing and general-purpose embedded development. The design is inspired by the functionality and general architecture of the STM32 NUCLEO-F401RE while requiring an independently created schematic and PCB layout. The board must integrate the MCU, regulated power, programming/debug access, user controls and commonly used communication interfaces within a compact two-layer PCB.

2. Proposed Solution

A custom development board centered on the STM32F401RET6 ARM Cortex-M4 microcontroller is proposed. USB Micro-B provides the primary 5 V input and USB data connection. The input passes through a resettable polyfuse and is converted to a regulated 3.3 V rail using an LD1117S33CTR linear regulator. The MCU is programmed and debugged through an external ST-LINK using a dedicated SWD header, avoiding the additional circuitry and board area required for an onboard debugger.

The board also exposes UART, SPI and I²C interfaces and provides GPIO expansion, a user push button, reset control, power and user LEDs and an external 8 MHz crystal. This creates a compact platform suitable for firmware development, peripheral experiments and embedded-system prototyping.

3. Circuit Architecture

Power: USB VBUS (5 V) -- resettable polyfuse -- 3.3 V LDO -- common 3.3 V rail. Regulator input/output capacitors support stable operation, while dedicated decoupling capacitors are provided at the MCU supply pins.

MCU and clock: The STM32F401RET6 is the central processing device. An 8 MHz external crystal connected to the HSE oscillator pins provides the external high-speed clock source, with dedicated load capacitors.

Programming and debugging: PA13/SWDIO and PA14/SWCLK are routed to a 5-pin SWD header together with target 3.3 V, NRST and GND for connection to an external ST-LINK debugger.

USB and peripherals: USB D⁻ and D⁺ are connected to PA11 and PA12 respectively. UART uses PA2/PA3, SPI uses PA5/PA6/PA7 with a chip-select connection, and I²C uses PB6/PB7 with external 4.7 k Ω pull-ups. GPIO expansion exposes PC2-PC9 together with 3.3 V and GND (referred datasheet).

4. PCB Design

The PCB is implemented as a two-layer board with the STM32F401RET6 placed near the center and interface headers arranged around the board for accessibility. The USB connector and power-regulation section are placed together to keep the power path compact. The SWD header is

positioned for connection to an external programmer/debugger, while UART, SPI, I²C and GPIO headers provide organized peripheral access.

The bottom copper layer is used substantially as a GND plane to provide a low-impedance return path and support signal integrity. Local decoupling components are placed close to the MCU supply pins. Routing was performed with a 0.20 mm track width and 0.20 mm clearance, with compact 0.60 mm vias (0.30 mm drill) where required. Silkscreen labels identify the major interfaces, controls and GPIO pins to improve usability.

5. Major Components

Component	Function
STM32F401RET6	Main 32-bit ARM Cortex-M4 MCU, 84 MHz, 512 KB Flash, 96 KB SRAM, 64pins
LD1117S33CTR	3.3 V, 800 mA linear regulator
Micro-USB B connector	USB power input and USB D+/D– interface
Resettable polyfuse	USB input overcurrent protection
8 MHz crystal	External clock source
C&K PTS647SN38SMTR2-LFS	Reset and user push buttons
SWD / UART / SPI / I ² C / GPIO headers	Programming, debugging and peripheral expansion
Power & user LEDs	Power indication and GPIO-controlled visual output
Decoupling capacitors	Supply filtering, VCAP support and crystal loading

6. Design Considerations

- Power : The 3.3 V rail is regulated from USB input and supported by local bulk and high-frequency decoupling.
- MCU supply requirements: The STM32F401RET6 supply, VCAP1 and analog/reference decoupling follow the device hardware recommendations.
- Clock integrity: The 8 MHz crystal and its load capacitors are kept close to the oscillator pins and routed with short connections.
- Debug accessibility: SWD signals and target-voltage reference are exposed on a dedicated header for external ST-LINK use.
- Signal integrity and manufacturability: Controlled routing rules, a continuous GND plane, appropriate clearances and compact via dimensions were used.
- Usability: Interface headers and silkscreen labels are arranged to make common development connections easy to identify. For the communication interfaces, the symbol ‘•’ is provided to pin 1 to identify the orientation.
- Compactness: Functional blocks are placed to minimize unnecessary routing while maintaining accessibility and electrical separation.

7. Testing / Validation

Validation was carried out at the design level using schematic connectivity checks and PCB design-rule verification. Electrical Rule Check (ERC) was used to identify incorrect or incomplete electrical connections, while Design Rule Check (DRC) was used to verify track width, copper clearance, board-edge clearance, vias and other PCB constraints. The final ERC/DRC reports are provided separately with the project submission. Functional connections for USB, SWD, reset, clock, power and peripheral headers were also reviewed against the intended circuit architecture and

STM32F401RE device requirements.

As the project submission is based on PCB design, validation in this document refers to schematic and layout verification rather than physical prototype measurements.

8. Challenges

The main design challenges were fitting the required functionality into a compact two-layer board, maintaining adequate clearance around the LQFP64 MCU, routing multiple peripheral interfaces and providing a practical ground-plane arrangement. Additional care was required for the MCU power and VCAP connections, USB routing, oscillator placement and external SWD accessibility. PCB rule violations encountered during layout were corrected by adjusting routing and clearances while retaining the intended functional architecture.

9. Final Outcome

The resulting design is a compact STM32F401RET6 development board with the size 46*45mm that combines regulated 3.3 V power, USB connectivity, external SWD programming/debugging, clock circuitry, reset and user controls, status indication and UART, SPI, I²C and GPIO expansion in a two-layer PCB. The design is organized for practical embedded development while retaining a compact form and clear user-facing interface labeling. The completed schematic, PCB layout, BOM, verification reports and visual renders are also provided in the repository.